

VINAY RAWAT

vinay.rawat058@gmail.com | 812-320-4874 | [linkedin.com/in/vinayrawat](https://www.linkedin.com/in/vinayrawat) | github.com/vinayrawat058

EDUCATION

Indiana University Bloomington

Aug. 2024 - May 2026

Master of Science in Computer Science — **GPA 3.6/4**

Indiana, United States

- **Relevant Coursework:** Applied Algorithms, Software Engineering, Applied Machine Learning, Engineering Cloud Computing, Data Mining, Data Visualization

TECHNICAL SKILLS

Programming Languages: Java, Python, TypeScript, Postgres, Oracle SQL

Frameworks & Technologies: Spring Boot, RESTful APIs, Kafka, JSON, HTML, CSS, Bootstrap, JUnit, Swagger

Developer Tools: Development & Collaboration (Git, GitHub, Postman), IDE (Visual Studio Code, Eclipse, IntelliJ).

Cloud and DevOps: Docker, Kubernetes, Cloud Technologies (JetStream2, Oracle Cloud Infrastructure Foundation)

Methodologies: Agile - Sprints, Stand-ups, Retrospectives

WORK EXPERIENCE

PTC Inc.

July 2022 – July 2024

Software Engineer (Full-stack development, Java 8+, Spring Boot, Postgres, TypeScript, Git, Docker)

- Designed high-performance **REST APIs in Spring Boot** for the One Codebeamer **enterprise-grade software platform**, incorporating **efficient database queries** and data processing techniques, resulting in a **75% improvement** in system responsiveness and faster user interactions.
- **Developed Java infrastructure** for PTC RV&S to enable the conversion of tags from HTML5 to legacy format, handling the deprecation of the **legacy output plugin** previously provided by TinyMCE while ensuring **prevention of data loss** for existing user information and seamless handling of new inputs.
- **Engineered seamless data exchange** between RESTful APIs and Angular frontend; **crafted 200+ JUnit test cases**, fortifying the system against edge cases and errors, while improving code maintainability by 15%.
- Resolved 40+ critical **production issues** for automotive clients by debugging **Spring services and Dockerized environments** while collaborating cross-functionally to deliver 24-hour hotfixes, demonstrating strong problem-solving and **ability to collaborate** while **maintaining 99.9% uptime** in high pressure settings.

PTC Inc.

Aug. 2021 – June 2022

Intern (Python, Oracle, Git)

- Studied and executed **15+ complex SQL scripts** (Oracle) to validate data integrity during database migrations, ensuring zero discrepancies and seamless transitions across environments.
- Optimized database **backup processes by 80% through Python** scripting and query tuning, enhancing system efficiency and reliability in cloud-native services showing ability to make decisions and work independently.

PROJECTS

Multi-Node Cloud Architecture for Real-Time Smart Home Insights

- Architected a cloud-deployed, fault-tolerant IoT analytics pipeline across **3 Ubuntu VMs on Jetstream**, leveraging **Kafka (multi-broker)** and Spark Streaming for real-time processing and anomaly detection.
- Designed tiered storage with **HDFS (Parquet, 100K records)** and PostgreSQL for **low-latency queries**, configured Grafana (10s refresh) for live monitoring and achieved fault tolerance via Kafka replication and HDFS checkpointing.

Scalable Payment Enabled Event Hosting Platform

- Engineered a **scalable fintech-style event platform** using Python/Django and SQLite, enabling 500+ users to manage event logistics and **financial data through a secure, intuitive dashboard**, improved process efficiency by 40%.
- Integrated secure payment workflows using APIs and deployed on Heroku, **maintaining 99% uptime for 1,500+ transactions** while reducing backend costs by 30% through resource optimization.

CORPORATE RECOGNITIONS

- **PTC Inc. Brains, Passion and Fun Award:** Recognized for rapidly implementing key features in the PTC RV&S web product.
- **PTC Inc. Customer First Award:** Awarded for swiftly unblocking a pivotal customer issue by debugging and delivering an effective quick fix.